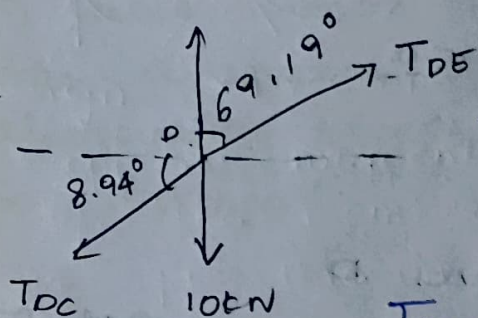


Consider f.B.D of point D.



$$\sum F_x = 0 \quad (1)$$

$$T_{DE} \sin 69.19 - T_{DC} \cos 8.94 = 0$$

$$\sum F_y = 0$$

$$T_{DE} \cos 69.19 - T_{DC} \sin 8.94 - 10 = 0$$

$$T_{DE} = 48.02 \text{ kN}$$

$$T_{DC} = 44.93 \text{ kN}$$

Since Tension in the segment AB is maximum, therefore maximum tension can be calculated at Support A.

$$T_{max} = \sqrt{V_A^2 + H^2}$$

$$= \sqrt{20^2 + 44.8^2}$$

$$= 48.78 \text{ kN}$$

Size of the cable

$$\sigma = 150 \text{ mpa}$$

$$= 150 \text{ N/mm}^2$$

$$\sigma = \frac{T_{max}}{\frac{\pi d^2}{4}}$$

$$150 = \frac{48.78 \times 10^3}{\frac{\pi d^2}{4}}$$

$$d = 20.34 \text{ mm}$$

ANALYSIS OF INDETERMINANT STRUCTURES.Slope-Deflection Method.

force -

There are two methods of analysis.

1. force method - flexibility [DS]
2. Displacement method - stiffness method. [DK - DOF]

Degree's of freedom [DOF]

$$n = n_{JK}$$

$$DOF = n - J \quad nJ - K$$

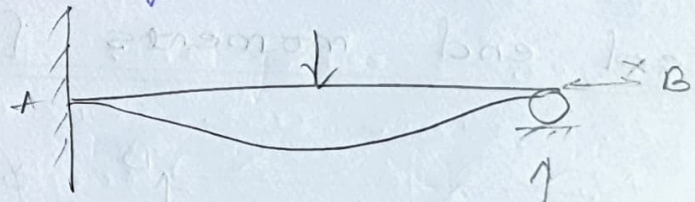
When a structure is loaded specified points on it called nodes or joints, will undergo unknown displacements. These displacements are referred as, the degree of freedom for the structure, and it is very important to specify DOF in the displacement method of analysis.

Since, displacements are unknowns in the displacement method of analysis, the number of these unknowns is referred as the degree in which the structure is kinematically indeterminate.

eg., for a beam structure only deformation due to bending is considered then the linear displacement only the axis of the beam is ignored.

Neglect axial deformation Δ .

$$DOF = 1 = 0$$



$$\text{Joint } A - 0$$

$$\text{Joint } B - 2$$

2

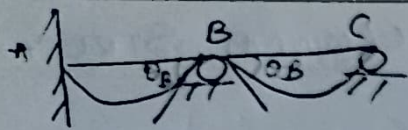
but horizontal is negligible.

$$\text{Joint } A - 0$$

$$\text{Joint } B - 1$$

1

$$DOF, DK = [1]$$



DOF = 3

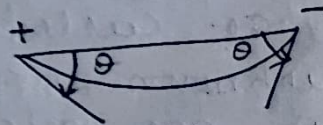
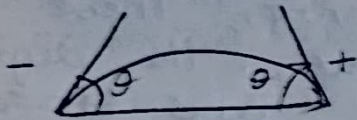
Sign Conventions.

1. Support moment of the member is positive (ve)

or moment at the end which is acting clockwise



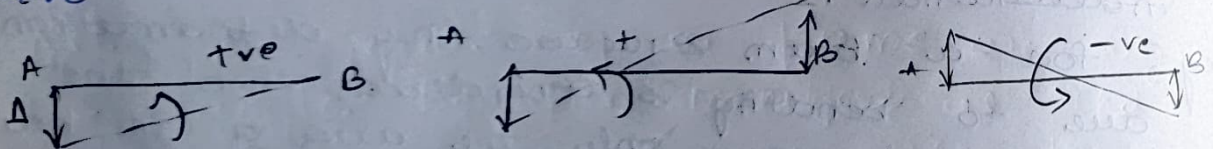
2. clockwise slope or angular rotation θ is positive



3. Displacement of right end w.r.t left end is Δ is positive when the beams moves upward.

(if member moves on counter clockwise its +ve)

Counter clockwise rotation of member is +ve



fixed end moments [FEM]

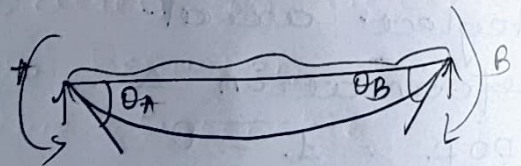
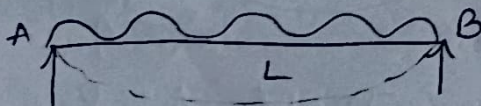
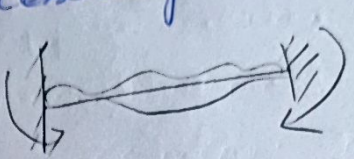


Fig 1 Shows simply supported beam with general loading. As beam bends, Slopes θ_A and θ_B occurs at supports A

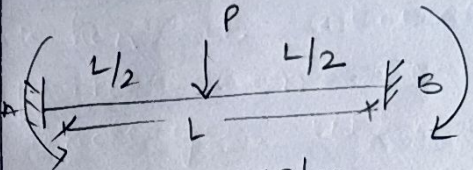
and B.

To bring back beam to original position it is necessary to apply end moments. M_A and M_B

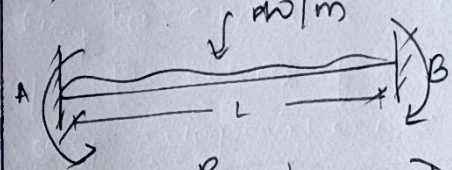


($\oplus$ ve) ($\ominus$ ve)

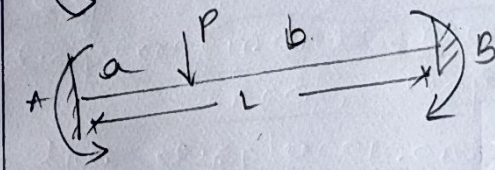
In case of fixed beam, end moments automatically develop as slopes at zero. Hence, end moments which acquire to keep the slope zero are called as fixed end moments [FEM]



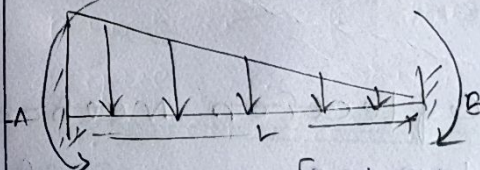
$$M_{FAB} = -\frac{PL}{8} \quad M_{BA} = +\frac{PL}{8}$$



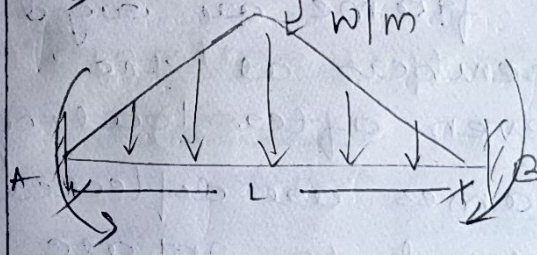
$$M_{AB} = -\frac{wL^2}{12} \quad M_{BA} = +\frac{wL^2}{12}$$



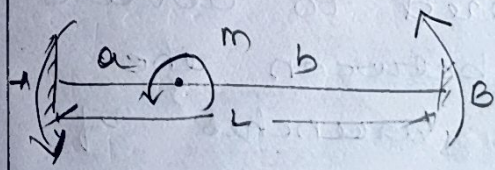
$$M_{AB} = \frac{P \times a b^2}{L^2} \quad M_{BA} = +\frac{P \times a^2 b}{L^2}$$



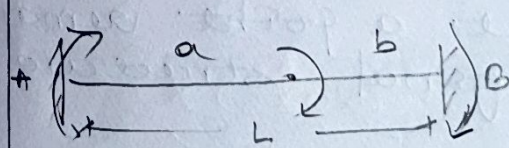
$$M_{AB} = -\frac{wL^2}{30} \quad M_{BA} = +\frac{wL^2}{30}$$



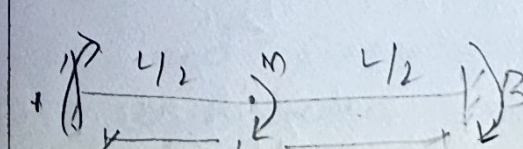
$$M_{AB} = -\frac{5}{96} wL^2 \quad M_{BA} = +\frac{5}{96} wL^2$$



$$M_{AB} = -\frac{mb}{L^2} (b-2a) \quad M_{BA} = +\frac{mb}{L^2} (b-2a)$$



$$M_{AB} = +\frac{mb}{L^2} (b-2a) \quad M_{BA} = +\frac{mb}{L^2} (b-2a)$$



$$M_{AB} = \frac{mb}{L^2} \left(\frac{b-2a}{2} \right) \quad M_{BA} = \frac{mb}{L^2} \left(\frac{b-2a}{2} \right)$$

$$\frac{ma(2b-a)}{L^2}$$

$$a=b$$

$$\frac{ma(2a-a)}{L^2}$$

$$\frac{ma(a)}{L^2}$$

$$\frac{m \times \frac{L}{2} \times \frac{L}{2}}{L^2}$$

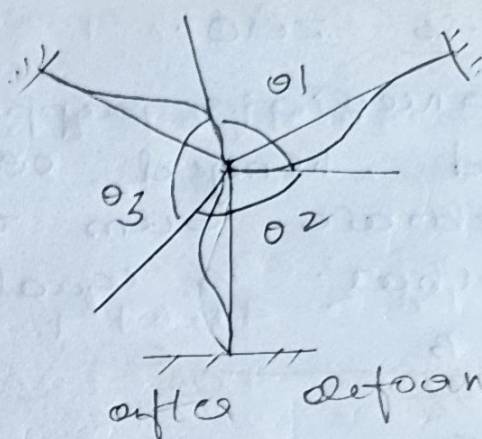
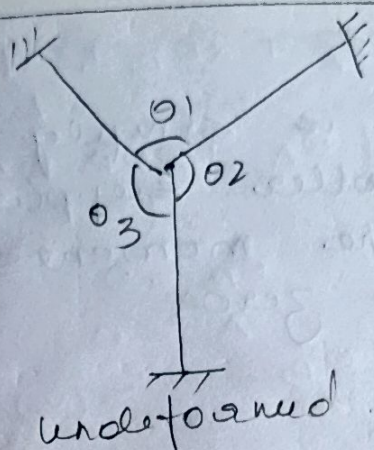
$$\frac{m \times \frac{L^2}{4}}{L^2}$$

$$\frac{m}{4}$$

15-07-2024

Assumptions in Slope deflection method

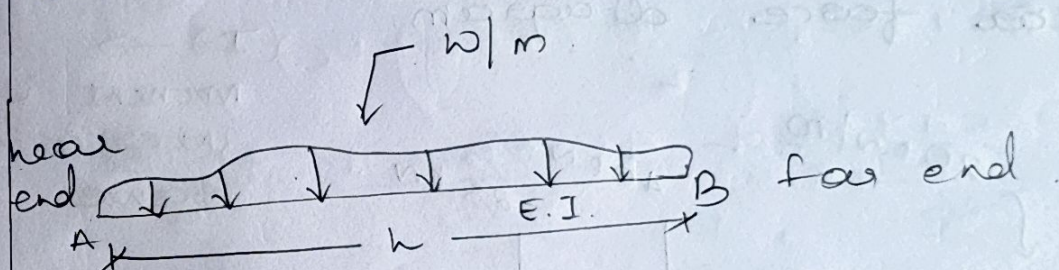
1. All the joints of a frame are rigid i.e., angle between members at the joint do not change even after loading.
2. Whenever beams and frames are deflected rigid joints are considered to rotate as a whole. i.e., angle between the tangents to the various branches of elastic curve meeting at a joint remains same as those in original structure.



[angle
does not
change]

3. Shear and axial deformations are negligible General Equation of Slope Deflection.

Consider a member AB of span L flexural rigidity EI subjected to general loading as shown in figure.



$$M_{AB} = M_{FAB} + \frac{2EI}{L} \left(2\theta_A + \theta_B + \frac{3\Delta}{L} \right)$$

$$M_{BA} = M_{FBA} + \frac{2EI}{L} \left(\theta_A + 2\theta_B + \frac{3\Delta}{L} \right)$$

Boundary Conditions and equilibrium equations.

Slope deflection equations are not sufficient to determine the unknowns.

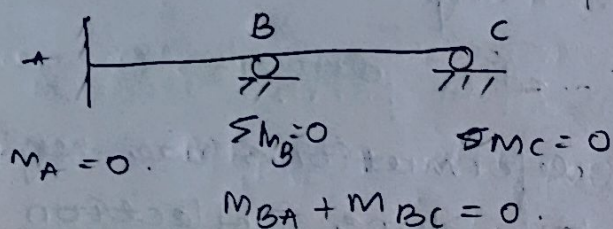
Hence, boundary conditions and equilibrium equations along with slope deflection equations can give the unknowns.

1. If the support is fixed, $\theta = 0$. Slope at θ is zero.

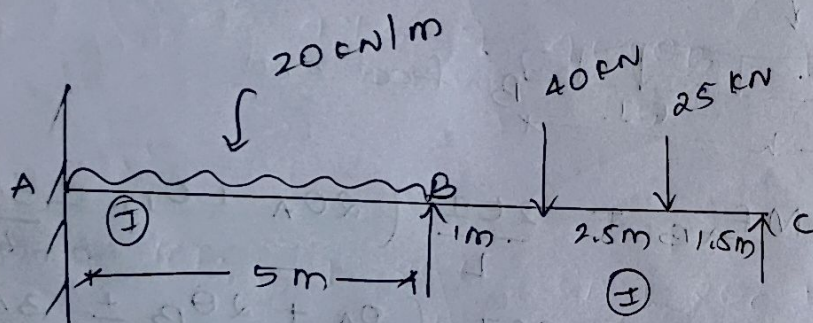
2. If end supports are hinged or simply supported or roller, then the final support

moment is zero.

3. If intermediate support is simply supported, hinged or roller support then algebraic sum of the moments about that is equal to zero.



1. Analyse the continuous beam shown in figure using slope deflection method and draw bending moment diagram and shear force diagram.



(I) →
moment of
interior is
same

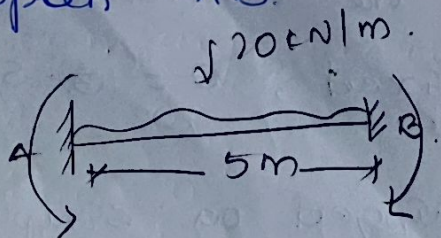
Step 1:

The degrees of freedom of for the given problem is, $DOF = 2 (\theta_B, \theta_C)$

Step 2:

fixed end moments.

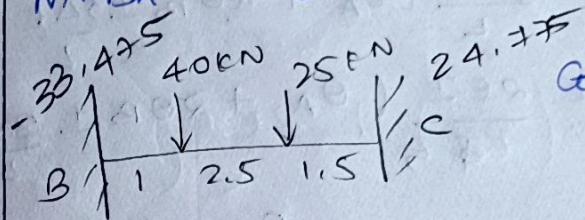
Span AB



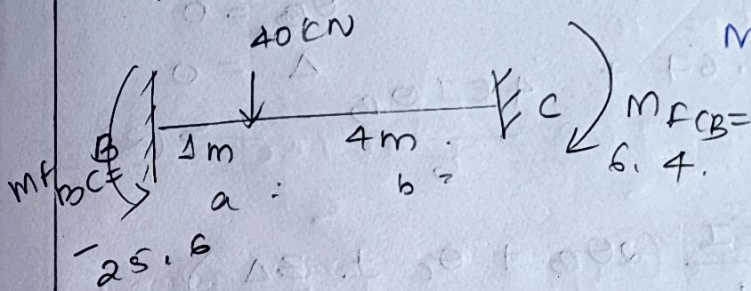
$$M_{FAB} = - \frac{wL^2}{12}$$

$$= - \frac{(20 \times 5) \times 5^2}{12}$$

$$= - 208.33 \text{ kN-m}$$

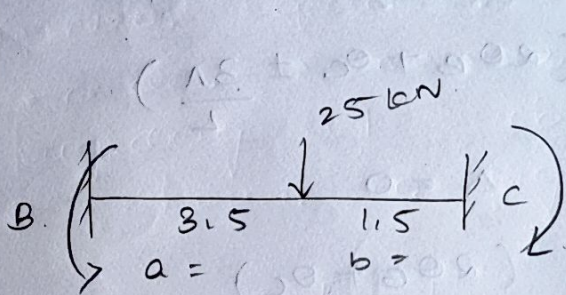
$$MFBA = 208.33 \text{ cm-m}$$


Consider principle of
Superposition.



$$M_{BC} = - \frac{P \times ab^2}{L^2}$$

$$= - 25.6 \text{ kN-m.}$$



$$m_{CB} = \frac{p \times a^2 b}{L^2}$$
$$= \frac{25.6 \text{ kN-m}}{6.4}$$

$$M_{BC}^A = - \frac{P \times a \times b^2}{L^2} = - \frac{25 \times 3.5 \times 1.5^2}{5^2} = -7.875 \text{ kN-m.}$$

$$M_{fcb} = \frac{p \times a^2 b}{5^2} = \frac{25 \times 3.5^2 \times 1.5}{5^2} = 18.375 \text{ kNm}$$

step 3:

slope deflection equation.

Span AB

$$M_{AB} = M_{FAB} + \frac{2EI}{L} (2\theta_A + \theta_B \pm \frac{3\Delta}{L})$$

Since support A is fixed $\therefore \theta_A = 0$
 $\Delta = 0$

$$M_{AB} = -41.67 + \frac{2EI}{5} \theta_B \rightarrow (1)$$

$$M_{BC} \neq M_{BA} = M_{F_{BA}} + \frac{2EI}{L} (\theta_A + 2\theta_B \pm \frac{3\Delta}{L})$$

$$= 41.67 + \frac{2EI}{5} (\theta_A + 2\theta_B \pm \frac{3\Delta}{L})$$

$$M_{BA} = 41.67 + \frac{4EI\theta_B}{5} \quad \begin{matrix} \theta_A = 0 \\ \Delta = 0 \end{matrix} \rightarrow (2)$$

Span B C

$$M_{BC} = M_{F_{BC}} + \frac{2EI}{L} (2\theta_B + \theta_C \pm \frac{3\Delta}{L})$$

$$= -\cancel{7.875} - 33.475 + \frac{2EI}{L} (2\theta_B + \theta_C \pm \frac{3\Delta}{L})$$

$$\Delta = 0$$

$$= -33.475 + \frac{2EI}{5} (2\theta_B + \theta_C)$$

$$= -33.475 + \frac{4EI\theta_B}{5} + \frac{2EI\theta_C}{5}$$

$$M_{CB} = M_{F_{CB}} + \frac{2EI}{L} (\theta_B + 2\theta_C \pm \frac{3\Delta}{L}) \rightarrow (3)$$

$$= 24.775 + \frac{2EI}{L} (\theta_B + 2\theta_C)$$

$$= 24.775 + \frac{2EI\theta_B}{5} + \frac{4EI\theta_C}{5} \rightarrow (4)$$

Step 4:

Boundary Conditions.

$$M_C = 0$$

$$M_{CB} = 0$$

$$i.e. \frac{2EI\theta_B}{5} + \frac{4EI\theta_C}{5} = -24.775 \rightarrow (5)$$

$$EM_B = 0$$

add (2) + (3)

$$41.67 + \frac{4EI\theta_B}{5} + (-33.445) + \frac{4EI\theta_B}{5} + \frac{2EI\theta_C}{5}$$

$$\frac{4EI\theta_B}{5} + \frac{4EI\theta_B}{5} + \frac{2EI\theta_C}{5} = -41.67 + 33.44$$

$$\frac{8}{5}EI\theta_B + \frac{2}{5}EI\theta_C = -8.195 \rightarrow (6)$$

solving equation (5) and (6)

$$EI\theta_B = 9.829 \quad 2.99 \sim 3$$

$$EI\theta_C = -32.466$$

$$\left\langle \theta_B = \frac{3}{EI} \right\rangle \left\langle \theta_C = -\frac{32.47}{EI} \right\rangle$$

Step 4

Final moments

By Substituting the values of θ_B and θ_C in equations 1, 2, 3 and 4

equ (1)

$$-41.67 + \frac{2EI}{5} \left(\frac{3}{EI} \right) = 0$$

$$M_{AB} = -40.47 \text{ kN-m}$$

equ (2)

$$M_{BA} = 41.67 + \frac{4EI}{5} \left(\frac{3}{EI} \right)$$

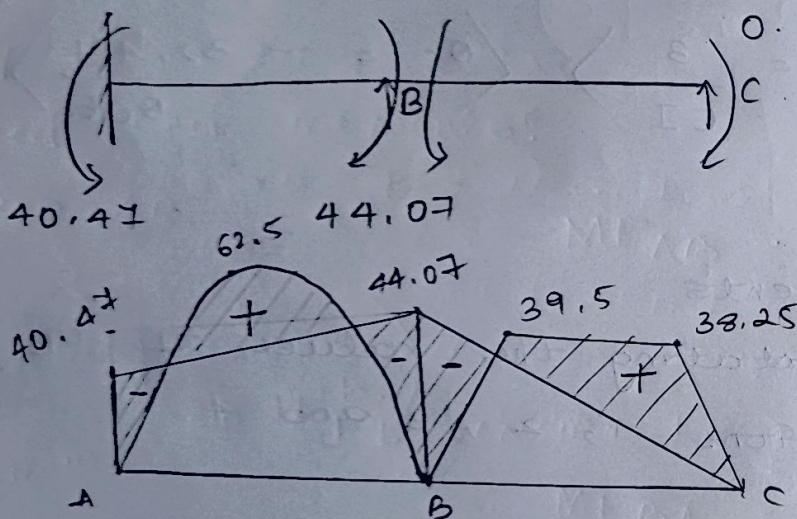
$$= 44.07 \text{ kN-m}$$

$$\begin{aligned}
 M_{BC} &= -33.475 + \frac{4EI}{5} \theta_B + \frac{2EI}{5} \theta_C \\
 &= -33.475 + \frac{4EI}{5} \left(\frac{3}{EI} \right) + \frac{2EI}{5} \left(-\frac{32.47}{EI} \right)
 \end{aligned}$$

$$M_{BC} = -44.063 \text{ kN-m.}$$

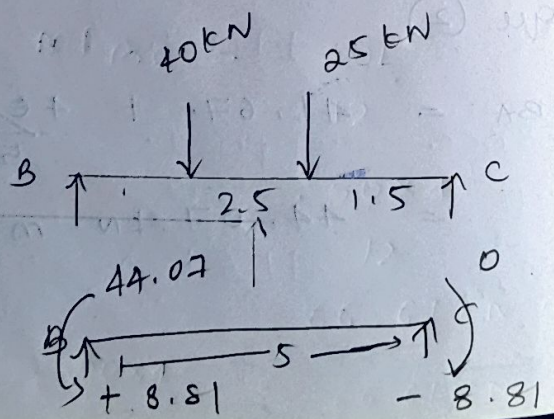
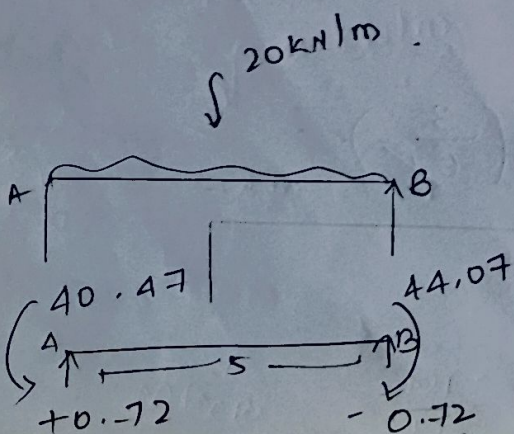
$$\begin{aligned}
 M_{CB} &= 24.975 + \frac{2EI}{5} \theta_B + \frac{4EI}{5} \theta_C \\
 &= 24.975 + \frac{2EI}{5} \left(\frac{3}{EI} \right) + \frac{4EI}{5} \left(-\frac{32.47}{EI} \right)
 \end{aligned}$$

$$M_{CB} = -1 \times 10^{-3} \text{ kN-m.} \quad (\text{S.O.9})$$



Bending moment of UDL = $\frac{wL^2}{8} = 62.5$

Construction of shear force diagram.



$$\sum M_A = 0$$

$$(20 \times 5) \times 2.5 - V_B \times 5$$

$$V_B = 50 \text{ kN}$$

$$V_A = V_B = 50 \text{ kN}$$

u/y

$$\sum M_B = 0$$

$$(40 \times 1) + (25 \times 3.5) - V_C \times 5 = 0$$

$$V_C = 25.5 \text{ kN}$$

$$\sum f_y = 0$$

$$V_B + V_C = 65 \text{ kN}$$

$$V_B = 39.5 \text{ kN}$$

$$\sum M_A = 0; -40.47 + 44.07 - V_B \times 5$$

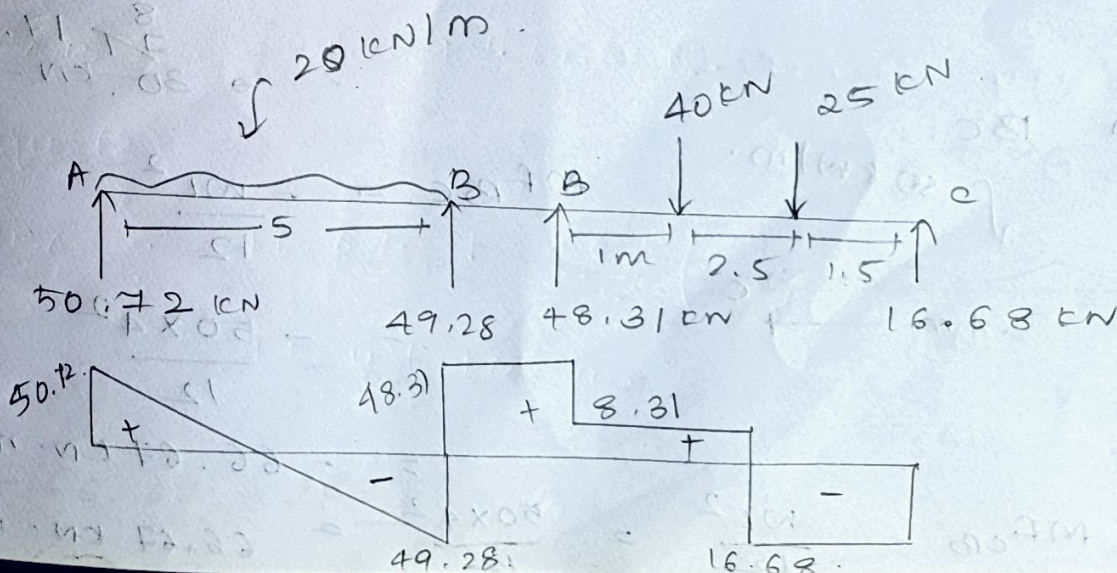
$$V_A = 0.72 \text{ kN}$$

$$V_B = -0.72 \text{ kN}$$

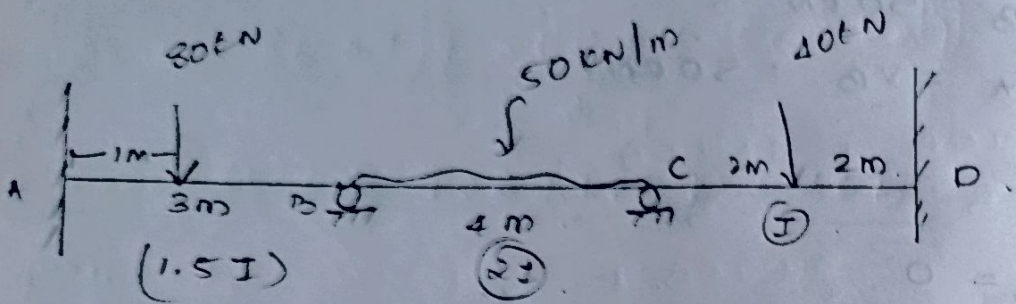
$$\sum M_B = 0; -44.07 - V_C \times 3.51$$

$$V_B = 8.81 \text{ kN}$$

$$V_C = -8.81 \text{ kN}$$



2. Analyse the continuous beam shown in figure using slope deflection method. Draw SFD and BMD.

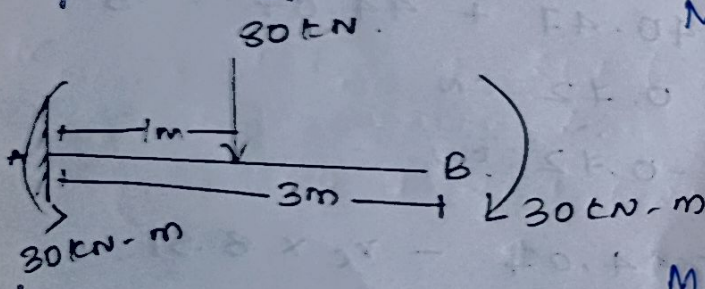


The degrees of freedom for the given problem is DOF = 2
(i.e., θ_B and θ_C).

Step 1

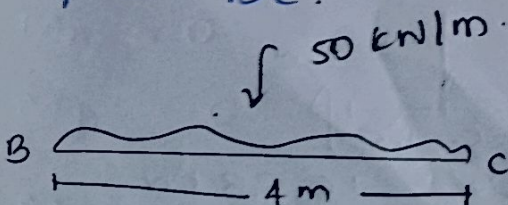
fixed End moments.

Span AB.



$$\begin{aligned}
 M_{FAB} &= \frac{PL}{8} - \frac{Pab^2}{L^2} \\
 &= \frac{80 \times 3}{8} - \frac{30 \times 2^2}{3^2} \\
 &= 30 \text{ kN-m} \\
 M_{FBA} &= -\frac{PL}{8} + \frac{Pa^2b}{L^2} \\
 &= -\frac{80 \times 3}{8} + \frac{30 \times 1^2}{3^2} \\
 &= -17.78 \text{ kN-m}
 \end{aligned}$$

Span BC.



$$\begin{aligned}
 M_{FBC} &= -\frac{WL^2}{12} \\
 &= -\frac{50 \times 4^2}{12} \\
 &= -66.67 \text{ kN-m} \\
 M_{FCB} &= \frac{WL^2}{12} = \frac{50 \times 4^2}{12} = 66.67 \text{ kN-m}
 \end{aligned}$$

Span CD.

$$M_{FCD} = -\frac{PL}{8} = -\frac{40 \times 4}{8} = -20 \text{ kN-m}$$

$$M_{FDC} = \frac{PL}{8} = \frac{40 \times 4}{8} = +20 \text{ kN-m}$$

Step 2

Slope deflection equation.

Span AB.

$$\begin{aligned} M_{AB} &= M_{FAB} + \frac{2EI}{L} (2\theta_A + \theta_B \pm \frac{3\Delta}{L}) \\ &= -35.56 + \frac{2EI}{L} (\theta_B) \end{aligned}$$

$$\therefore \theta_A = 0$$

$$(v) = -35.56 + \frac{2EI\theta_B}{L}$$

$$\left(\frac{1}{EI} \right) (-35.56 + \frac{3\theta_B}{L}) \rightarrow (i)$$
$$= -35.56 + \frac{3EI\theta_B}{L} = (-35.56 + EI\theta_B)$$

$$M_{BA} = M_{FBA} + \frac{2EI}{L} (2\theta_B + \theta_A \pm \frac{3\Delta}{L})$$

$$(iv) = 17.78 + \frac{2EI}{L} (2\theta_B)$$

$$= 17.78 + \frac{4EI\theta_B}{L} \rightarrow (ii)?$$

$$[17.78 + \frac{2EI\theta_B}{L}]$$

$$M_{BC} = M_{FBC} + \frac{2EI}{L} (2\theta_B + \theta_C \pm \frac{3\Delta}{L})$$

$$= -66.67 + \frac{2EI}{L} (2\theta_B + \theta_C)$$

$$= -66.67 + \frac{4EI\theta_B}{L} + \frac{2EI\theta_C}{L}$$

$$= -66.67 + \frac{2EI\theta_B}{L} + \frac{4EI\theta_C}{L} \rightarrow (iii)$$

$$M_{CB} = M_{FCB} + \frac{2EI}{L} (2\theta_C + \theta_B + \frac{3\Delta}{L})$$

$$= 66.67 + \frac{2EI}{L} (2\theta_C + \theta_B)$$

$$= 66.67 + \frac{2 \times 21}{4} (2\theta_C + \theta_B)$$

$$= 66.67 + 21\theta_C + 10.5\theta_B \rightarrow (iv)$$

Span CD

$$M_{CD} = M_{FCD} + \frac{2EI}{L} (2\theta_C + \theta_D + \frac{3\Delta}{L})$$

$$= -20 + \frac{2EI}{4} (2\theta_C + \theta_D)$$

$$= -20 + \frac{2 \times 21}{4} (2\theta_C + \theta_D)$$

$$= -20 + 21\theta_C + 10.5\theta_D \rightarrow (v)$$

$$M_{DC} = M_{FDC} + \frac{2EI}{L} (2\theta_D + \theta_C + \frac{3\Delta}{L})$$

$$= 20 + \frac{2EI}{4} (2\theta_D + \theta_C)$$

$$= 20 + 21\theta_D + 10.5\theta_C \rightarrow (vi)$$

$$= 20 + 0.5EI\theta_C$$

Step 3

Boundary Conditions

$$M_A = 0$$

$$M_D = 0$$

$$M_B = \Sigma M_B$$

$$M_C = \Sigma M_C$$

$$\Sigma M_{BC} = 6I\theta_B + 6I\theta_C = 66.67 \rightarrow (vi).$$

ΣM

$$\begin{aligned} M_A &= 0 \\ M_D &= 0 \end{aligned} \left\} \text{fixed at the end.}$$

$$\Sigma M_B = 0.$$

$$M_{BA} + M_{BC} = 0.$$

$$17.78 + 2EI\theta_B + (-66.67) + 2EI\theta_B + 6I\theta_C.$$

$$4EI\theta_B + 6I\theta_C = 48.89$$

$$\Sigma M_C = 0$$

$$M_{CB} + M_{CD} = 0.$$

$$66.67 + 2EI\theta_C + 6I\theta_B + (-20) + 6I\theta_C +$$

$$3EI\theta_C + 6I\theta_B = -46.67.$$

$$\theta_B = \frac{17.57}{EI}$$

$$\theta_C = -\frac{21.415}{EI}$$

Squ (1)

$$M_{AB} = -17.99 \approx 18 \text{ kN-m}$$

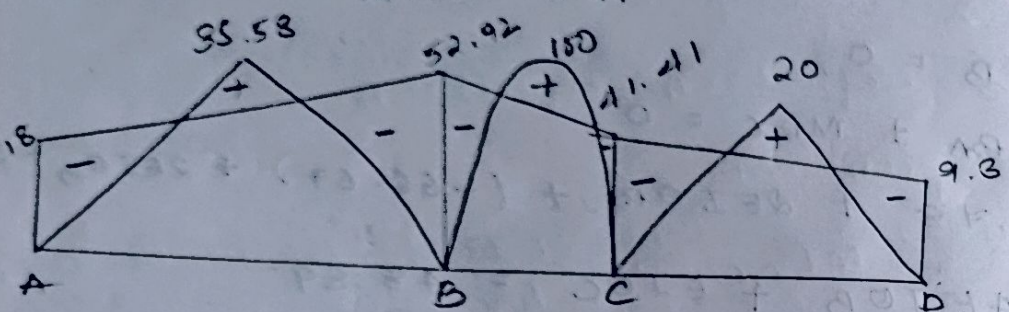
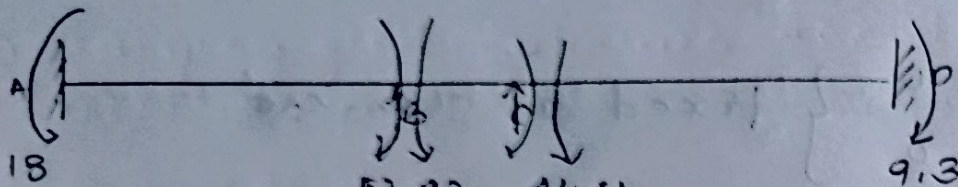
$$M_{BA} = 52.92 \text{ kN-m}$$

$$M_{BC} = -52.92 \text{ kN-m}$$

$$M_{CB} = 41.41 \text{ kN-m}$$

$$M_{CD} = -41.41 \text{ kN-m}$$

$$M_{DC} = 9.3 \text{ kN-m}$$



Construction of shear force diagram

$$\frac{18 \times 13}{13} = 18$$

$$\frac{211.36}{13} = 16.26$$

(F) up

$$m - 18 \times 13 = 18 \times 13$$

$$m - 18 \times 13 = 18 \times 13$$

$$m - 18 \times 13 = 18 \times 13$$

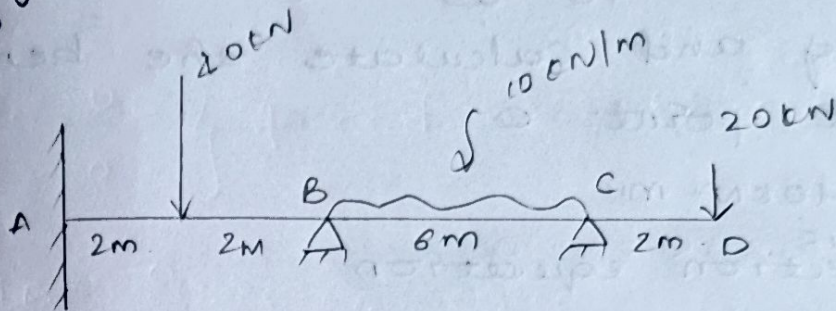
$$m - 18 \times 13 = 18 \times 13$$

$$m - 18 \times 13 = 18 \times 13$$

$$m - 18 \times 13 = 18 \times 13$$

$$m - 18 \times 13 = 18 \times 13$$

3. Analyse the continuous beam shown in figure using slope deflection method. Draw SFD and BMD



Degree of freedom, Dof is 2 (θ_B and θ_C)

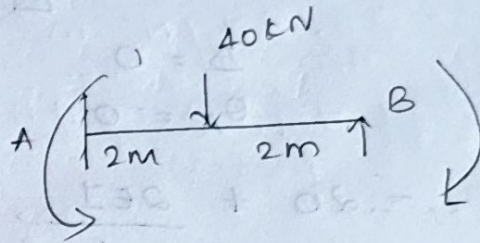
(i) fixed end moments

Span AB

$$M_{AB} = -\frac{PL}{8}$$

$$= -\frac{40 \times 4}{8}$$

$$= -20 \text{ kN-m}$$



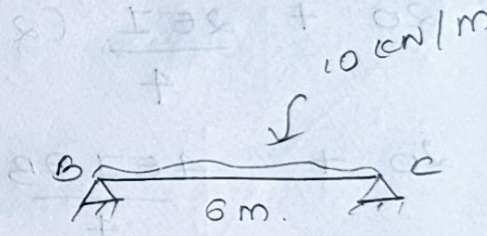
$$M_{BA} = +\frac{PL}{8} = \frac{40 \times 4}{8} = 20 \text{ kN-m}$$

Span BC

$$M_{BC} = -\frac{wL^2}{12}$$

$$= -\frac{10 \times 6^2}{12}$$

$$= -30 \text{ kN-m}$$



$$M_{CB} = \frac{wL^2}{12}$$

$$= \frac{10 \times 6^2}{12}$$

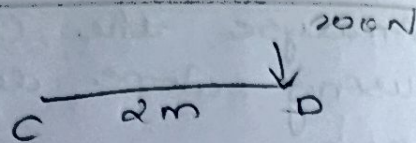
$$= 30 \text{ kN-m}$$

Span CD.

Consider portion CD as

overhanging and calculate the bending moment at point C

$$M_{CD} = -40 \text{ kN-m}$$



(i) Slope deflection equation

Span AB

$$M_{AB} = M_{FAB} + \frac{2EI}{L} (2\theta_A + \theta_B + \frac{3\Delta}{L})$$

$$\Delta = 0$$

$$\theta_A = 0$$

$$= -20 + \frac{2EI}{4} (\theta_B)$$

$$= -20 + \frac{EI\theta_B}{2} \rightarrow (i)$$

$$M_{BA} = M_{FBA} + \frac{2EI}{L} (2\theta_B + \theta_A + \frac{3\Delta}{L})$$

$$= 20 + \frac{2EI}{4} (2\theta_B)$$

$$= 20 + \frac{4EI\theta_B}{4}$$

$$= 20 + EI\theta_B \rightarrow (ii)$$

Span BC

$$M_{BC} = M_{FBC} + \frac{2EI}{L} (2\theta_B + \theta_C + \frac{3\Delta}{L})$$

$$= -30 + \frac{2EI}{6} (2\theta_B + \theta_C)$$

$$= -30 + \frac{4EI\theta_B}{6} + \frac{2EI\theta_C}{6}$$

$$= -30 + \frac{2EI\theta_B}{3} + \frac{EI\theta_C}{3} \rightarrow (iii)$$

$$M_{CB} = M_{FD} + \frac{2EI}{L} (2\theta_C + \theta_B + \frac{3\Delta}{L})$$

$$= 30 + \frac{2EI}{L} (2\theta_C + \theta_B)$$

$$= 30 + \frac{2EI}{16.146} (2\theta_C + \theta_B)$$

$$= 30 + \frac{2EI\theta_C}{3} + \frac{EI\theta_B}{3} \rightarrow (w)$$

ii) Boundary Conditions.

$$\Sigma M_B = 0$$

$$\Sigma M_C = 0$$

$$\Sigma M_B = 0; M_{BA} + M_{BC} = 0.$$

$$20 + EI\theta_B + \frac{2EI\theta_B}{3} + \frac{EI\theta_C}{3} - 30 = 0.$$

$$\frac{2}{3} EI\theta_B + \frac{EI\theta_C}{3} = 10 \rightarrow (a)$$

$$\Sigma M_C = 0; M_{CB} + M_{CD} = 0.$$

$$30 + \frac{2}{3} EI\theta_C + \frac{EI\theta_B}{3} - 40 = 0.$$

$$\frac{2}{3} EI\theta_C + \frac{EI\theta_B}{3} = 10 \rightarrow (b).$$

$$\theta_A = \frac{3.33}{EI}$$

$$\theta_C = \frac{13.33}{EI}$$

$$M_{AB} = -18.335 \text{ kN-m}$$

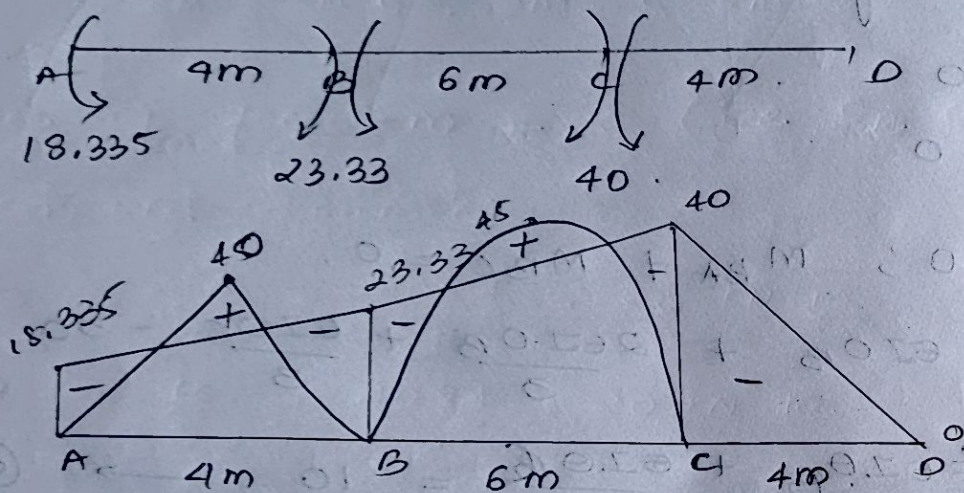
$$M_{BA} = 23.33 \text{ kN-m}$$

$$M_{BC} = -23.33 \text{ kN-m}$$

$$M_{CB} = 40 \text{ kN-m}$$

$$M_{CD} = -40 \text{ kN-m}$$

(iv) BMD.



(v) SFD.

Moments due to support settlement.

$$\begin{aligned}
 M_{AB} &= M_{FAB} + \frac{2EI}{L} (2\theta_A + \theta_B \pm \frac{3\Delta}{L}) \\
 &= M_{FAB} + \frac{6EI\Delta}{L^2} + \frac{2EI}{L} (2\theta_A + \theta_B) \\
 &= M_{FAB} + M'_{FAB} + \frac{2EI}{L} (2\theta_A + \theta_B).
 \end{aligned}$$

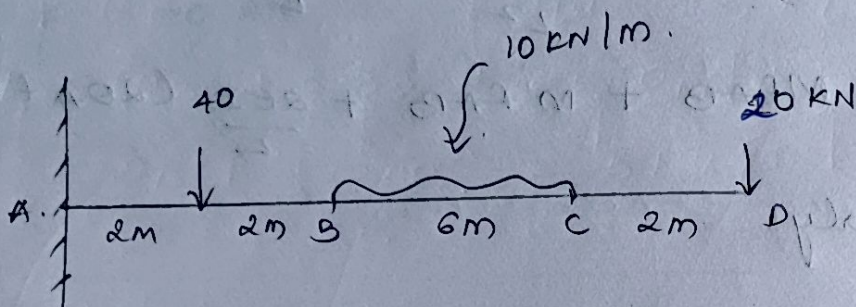
Similarly,

$$\begin{aligned}
 M_{BA} &= M_{FBA} + \frac{2EI}{L} (2\theta_B + \theta_A \pm \frac{3\Delta}{L}) \\
 &= M_{FBA} + \frac{6EI\Delta}{L^2} + \frac{2EI}{L} (\theta_A + 2\theta_B) \\
 &= M_{FBA} + M'_{FBA} + \frac{2EI}{L} (\theta_A + 2\theta_B).
 \end{aligned}$$

Problems on Support Settlement

4. Analyse the continuous beam shown in figure using slope deflection method. Support B sinks by 10mm.

Take $E = 200 \text{ GPa}$, $I = 100 \times 10^{-6} \text{ m}^4$.



Degree of freedom = 2 (i.e. θ_B and θ_C)

- (i) fixed end moments.

$$M_{FAB} = -\frac{PL}{8} = -\frac{40 \times 4}{8} = -20 \text{ kN-m}$$

$$M_{FBA} = \frac{PL}{8} = \frac{40 \times 4}{8} = 20 \text{ kN-m}$$

Span BC

$$M_{FBC} = -\frac{PL^2}{12} = -\frac{10 \times 6^2}{12} = -30 \text{ kN-m}$$

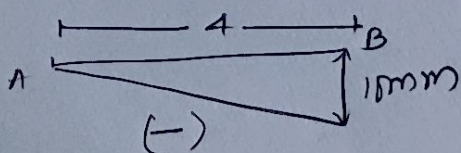
$$M_{FCB} = \frac{PL^2}{12} = \frac{10 \times 6^2}{12} = 30 \text{ kN-m}$$

Span CD

$$M_{CD} = -(20 \times 2) = -40 \text{ kN-m}$$

- (ii) Moments due to Support Settlement

Span AB.

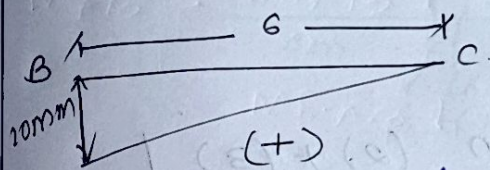


$$M'_{FAB} = M'_{FBA} = -\frac{6EI \times \Delta}{L^2}$$

$$= -\frac{6 \times 200 \times 10^6 \times 100 \times 10^{-6} \times 10 \times 10^{-3}}{4^2}$$

$$M'_{FAB} = M'_{FBA} = -75 \text{ kN-m.}$$

Span BC.



$$M'_{FBC} = M'_{FCB} = \frac{6EI}{L^2} \times \Delta$$

$$= \frac{6 \times 200 \times 10^6 \times 100 \times 10^{-6}}{6^2} \times 10 \times 10^{-3}$$

$$M'_{FBC} = M'_{FCB} = 33.33 \text{ kN-m.}$$

iii) Slope deflection equation.

Span AB

$$M_{AB} = M_{FAB} + M'_{FAB} + \frac{2EI}{L} (2\theta_A + \theta_B)$$

$$\theta_A = 0.$$

$$= -20 + (-75) + \frac{2EI}{4} (\theta_B)$$

$$= -95 + \frac{1}{2} EI \theta_B \rightarrow \textcircled{1}$$

$$M_{BA} = M_{FBA} + M'_{FBA} + \frac{2EI}{L} (2\theta_B)$$

$$= 20 - 75 + \frac{2EI}{4} (2\theta_B)$$

$$= -55 + EI \theta_B \rightarrow \textcircled{2}$$

Span BC.

$$M_{BC} = M_{FBC} + M'_{FBC} + \frac{2EI}{L} (2\theta_B + \theta_C)$$

$$= -30 + 33.33 + \frac{2EI}{6} (2\theta_B + \theta_C)$$

$$= 3.33 + \frac{2}{3} EI \theta_B + \frac{1}{3} EI \theta_C \rightarrow \textcircled{3}$$

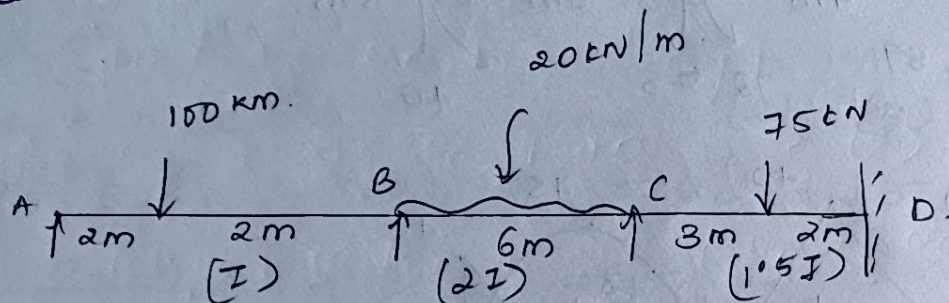
$$M_{CB} = M_{FCB} + M'_{FCB} + \frac{2EI}{L} (\theta_B + 2\theta_C)$$

$$= 30 + 33.33 + \frac{2EI}{6} (\theta_B + 2\theta_C)$$

$$= 63.33 + \frac{1}{3} EI \theta_B + \frac{2}{3} EI \theta_C \rightarrow \textcircled{4}$$

6. Analyse the continuous beam shown in figure using slope deflection method. Supports B sinks by 10mm and C rises by 5mm.

Take $EI = 2 \times 10^4 \text{ kN-m}^2$



Degree of freedom = 3 (i.e. θ_A , θ_B and θ_C)

(i) fixed end moments

Span AB

$$M_{FAB} = -\frac{PL}{8} = -\frac{100 \times 4}{8} = -50 \text{ kN-m}$$

$$M_{FBA} = \frac{PL}{8} = \frac{100 \times 4}{8} = 50 \text{ kN-m}$$

Span BC

$$M_{FBC} = -\frac{PL^2}{12} = -\frac{20 \times 6^2}{12} = -60 \text{ kN-m}$$

$$M_{FCB} = \frac{PL^2}{12} = \frac{20 \times 6^2}{12} = 60 \text{ kN-m}$$

Span CD

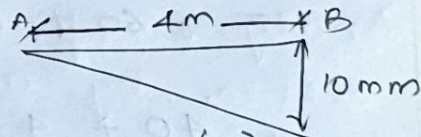
$$M_{FCD} = -\frac{Pab^2}{L^2} = -\frac{75 \times 3 \times 2^2}{5^2} = -36 \text{ kN-m}$$

$$M_{FDC} = \frac{Pa^2b}{L^2} = \frac{75 \times 3^2 \times 2}{5^2} = 54 \text{ kN-m}$$

ii) Moments due to Support Settlement
Span AB

$$m'_{FAB} = m'_{FBA} = - \frac{6EI}{L^2} \times \Delta$$

$$= - \frac{6 \times 2 \times 10^4}{4^2} \times 10 \times 10^{-3} \quad (-)$$



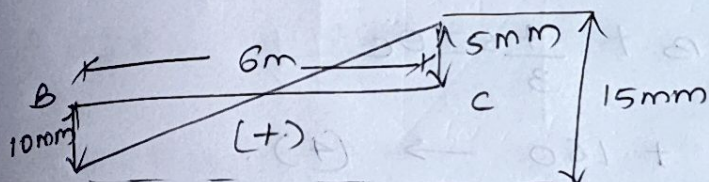
$$m'_{FAB} = m'_{FBA} = - 75 \text{ kN-m}$$

(25)

Span BC

$$m'_{BC} = m'_{CB} = \frac{6EI}{L^2} \times \Delta$$

$$= \frac{6 \times 2 \times 10^4}{6^2} \times 2 \times 15 \times 10^{-3}$$

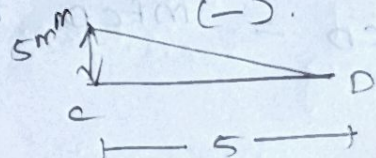


$$m'_{BC} = m'_{CB} = 100 \text{ kN-m}$$

Span CD

$$m'_{CD} = m'_{DC} = - \frac{6EI}{L^2} \times \Delta$$

$$= - \frac{6 \times 2 \times 10^4}{5^2} \times 1.5 \times 5 \times 10^{-5} \quad (-)$$



$$m'_{CD} = m'_{DC} = - 36 \text{ kN-m}$$

iii) Slope deflection method.

$$M_{AB} = M_{FAB} + m'_{FAB} + \frac{2EI}{L} (\theta_A + \theta_B)$$

$$= - 50 - 75 + EI\theta_A + \frac{1}{2}EI\theta_B$$

$$EI\theta_A + \frac{1}{2}EI\theta_B - 125 \rightarrow (i)$$

$$M_{BA} = M_{FBA} + m'_{FBA} + \frac{2EI}{L} (\theta_A + 2\theta_B)$$

$$= 50 - 75 + \frac{2EI}{4} (\theta_A + 2\theta_B)$$

$$= - 25 + \frac{1}{2}EI\theta_A + EI\theta_B$$

$$\frac{1}{2}EI\theta_A + EI\theta_B - 25 \rightarrow (ii)$$

Span BC

$$M_{BC} = M_{FBC} + M'_{FBC} + \frac{2EI}{L} (2\theta_B + \theta_C)$$

$$= -60 + 100 + \frac{2 \times 10^4 \times 1}{6} (2\theta_B + \theta_C)$$

$$= 40 + \frac{4}{3} EI \theta_B + \frac{2}{3} EI \theta_C$$

$$\frac{4}{3} EI \theta_B + \frac{2}{3} EI \theta_C + 40 \rightarrow (3)$$

$$M_{CB} = M_{FCB} + M'_{FCB} + \frac{2EI}{L} (\theta_B + 2\theta_C)$$

$$= 60 + 100 + \frac{2 \times 10^4 \times 1}{6} (\theta_B + 2\theta_C)$$

$$= 160 + \frac{2}{3} EI \theta_B + \frac{4}{3} EI \theta_C$$

$$\frac{2}{3} EI \theta_B + \frac{4}{3} EI \theta_C + 160 \rightarrow (4)$$

Span CD.

$$M_{CD} = M_{FCD} + M'_{FCD} + \frac{2EI}{L} (2\theta_C + \theta_D)$$

$$\theta_D = 0$$

$$= -36 - 36 + \frac{2 \times 10^4 \times 1.5}{5} (2\theta_C)$$

$$= -72 + \frac{6}{5} EI \theta_C$$

$$\frac{6}{5} EI \theta_C - 72 \rightarrow (5)$$

$$M_{DC} = M_{FDC} + M'_{FDC} + \frac{2EI}{L} (\theta_C)$$

$$= 54 - 36 + \frac{2 \times 10^4 \times 1.5}{5} (\theta_C)$$

$$= 18 + \frac{3}{5} EI \theta_C$$

$$\frac{3}{5} EI \theta_C + 18 \rightarrow (6)$$

(iv) Boundary conditions.

$$\Sigma M_B = 0$$

$$\Sigma M_C = 0$$

$$\theta_A = 0$$

$$\Sigma M_B = 0 ; M_{BA} + M_{BC} = 0.$$

$$\frac{1}{2} EI \theta_A + EI \theta_B - 25 + \frac{4}{3} EI \theta_B + \frac{2}{3} EI \theta_C + 40 = 0.$$

$$\frac{1}{2} EI \theta_A + \frac{7}{3} EI \theta_B + \frac{2}{3} EI \theta_C = -15 \rightarrow (a).$$

$$\Sigma M_C = 0 ; M_{CB} + M_{CD} = 0.$$

$$\frac{2}{3} EI \theta_B + \frac{4}{3} EI \theta_C + 160 + \frac{6}{5} EI \theta_C - 72 = 0.$$

$$\frac{2}{3} EI \theta_B + \frac{38}{15} EI \theta_C = -88 \rightarrow (b)$$

$$EI \theta_A + \frac{1}{2} EI \theta_B = 125 \rightarrow (c).$$

$$\theta_A = \frac{139.24}{EI}$$

$$\theta_B = \frac{-28.48}{EI}$$

$$\theta_C = \frac{-27.24}{EI}$$

$$M_{AB} = 0$$

$$M_{BA} = 16.14 \text{ kN-m}$$

$$M_{BC} = -16.14 \text{ kN-m}$$

$$M_{CB} = 104.69 \text{ kN-m}$$

$$M_{CD} = -104.69 \text{ kN-m}$$

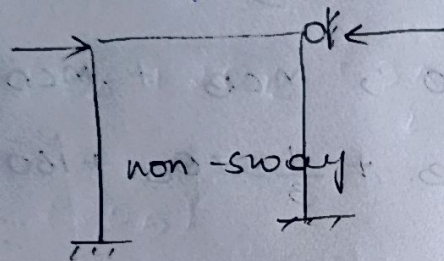
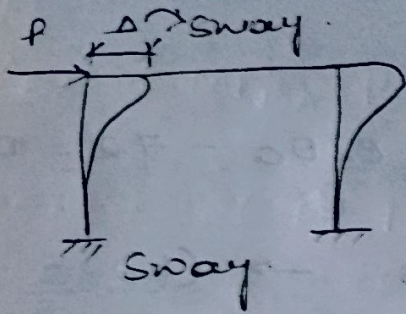
$$M_{DC} = 1.65 \text{ kN-m}.$$

Analysis of orthogonal rigid frames by slope deflection method.

Analysis of frames can be carried out either by a with sway, or without sway [called as non-sway]

What is sway?

The lateral movement of the top of the structure w.r.t the support.

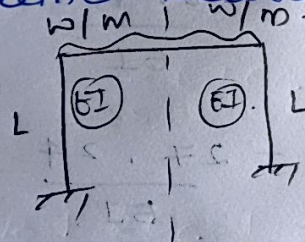


Condition for a non-sway analysis

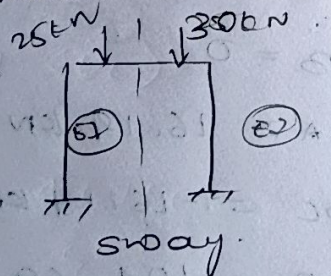
If the frame is having a support at the top, which produces horizontal reaction. They restrict sway movement.

If there is no horizontal force acting on the frame and frame satisfying all the four symmetrical condition which are

- (i) Material (E)
- (ii) Geometry (I)
- (iii) Loading (P)
- (iv) Support condition.

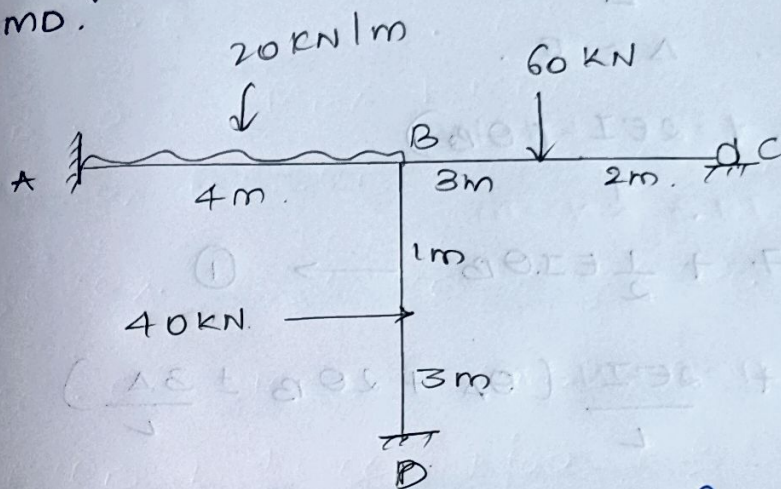


Symmetric non sway.



Analysis of non-sway frames.

Analyse the portal frame shown in figure using slope deflection method. Sketch BMD.



Degree of freedom $I = 2$ (i.e. θ_B and θ_C)

The given frame is non-sway because a fixed support at top (at A).

where $\theta_A = \theta_D = 0$.

Fixed end moments.

Span AB

$$M_{FAB} = -\frac{PL^2}{12} = -\frac{20 \times 4^2}{12} = -26.67 \text{ kN-m}$$

$$M_{FBA} = \frac{PL^2}{12} = \frac{20 \times 4^2}{12} = 26.67 \text{ kN-m}$$

Span BC

$$M_{FBC} = -\frac{Pab^2}{L^2} = -\frac{60 \times 3 \times 2^2}{5^2} = -28.8 \text{ kN-m}$$

$$M_{FCB} = \frac{Pa^2b}{L^2} = \frac{60 \times 3^2 \times 2}{5^2} = 43.2 \text{ kN-m}$$

Span DB

$$M_{FDB} = -\frac{Pab^2}{L^2} = -\frac{40 \times 3 \times 1^2}{4^2} = -7.5 \text{ kN-m}$$

$$M_{FBD} = \frac{Pab^2}{L^2} = \frac{40 \times 3 \times 1^2}{4^2} = 22.5 \text{ kN-m}$$

(ii) Slope deflection equation.

Span AB.

$$M_{AB} = M_{FAB} + \frac{2EI}{L} (\theta_A + \theta_B \pm \frac{3\Delta}{L})$$

$$\theta_A = 0, \Delta = 0.$$

$$= -26.67 + \frac{2EI}{4} (\theta_B)$$

$$= -26.67 + \frac{1}{2} EI \theta_B \rightarrow \textcircled{1}$$

$$M_{BA} = M_{FBA} + \frac{2EI}{L} (\theta_A + 2\theta_B \pm \frac{3\Delta}{L})$$

$$= 26.67 + \frac{2EI}{4} (2\theta_B)$$

$$= 26.67 + EI \theta_B \rightarrow \textcircled{2}$$

Span BC

$$M_{BC} = M_{FBC} + \frac{2EI}{L} (\theta_B + \theta_C \pm \frac{3\Delta}{L})$$

$$= -28.8 + \frac{2EI}{5} (\theta_B + \theta_C)$$

$$= -28.8 + \frac{4}{5} EI \theta_B + \frac{2}{5} EI \theta_C \rightarrow \textcircled{3}$$

$$M_{CB} = M_{FCB} + \frac{2EI}{L} (\theta_B + 2\theta_C \pm \frac{3\Delta}{L})$$

$$= 43.2 + \frac{2EI}{5} (\theta_B + 2\theta_C)$$

$$= 43.2 + \frac{2EI \theta_B}{5} + \frac{4EI \theta_C}{5} \rightarrow \textcircled{4}$$

Span DB.

$$M_{DB} = M_{FDB} + \frac{2EI}{L} (\theta_D + \theta_B \pm \frac{3\Delta}{L})$$

$$\theta_D = 0.$$

$$M_{DB} = m_{fDB} + \frac{2EI}{L} (\theta_B)$$

$$= -7.5 + \frac{2EI\theta_B}{4}$$

$$= -7.5 + \frac{1}{2}EI\theta_B \rightarrow (5)$$

$$M_{BD} = m_{fBD} + \frac{2EI}{L} (2\theta_B)$$

$$= 22.5 + \frac{2EI}{4} (2\theta_B)$$

$$= 22.5 + EI\theta_B \rightarrow (6)$$

1) Boundary conditions.

$$\Sigma M_B = 0;$$

$$M_{BA} + M_{BC} = 0.$$

$$26.67 + EI\theta_B - 28.8 + \frac{4}{5}EI\theta_B + \frac{2}{5}EI\theta_C = 0.$$

$$\frac{9}{5}EI\theta_B + \frac{2}{5}EI\theta_C = 2.13 \rightarrow (a)$$

$$\theta_C = 0.$$

$$M_{CB} = 0.$$

$$\frac{2EI\theta_B}{5} + \frac{4EI\theta_C}{5} = -43.2 \rightarrow (b)$$

$$\Sigma M_B = 0$$

$$\Sigma M_{BA} + M_{BC} + M_{BD} = 0.$$

$$26.67 + EI\theta_B - 28.8 + \frac{4}{5}EI\theta_B + \frac{2}{5}EI\theta_C + 22.5 + EI\theta_B.$$

$$\frac{14EI\theta_B}{5} + \frac{2EI\theta_C}{5} = 20.37 \rightarrow (a)$$

$$\frac{2EI\theta_B}{5} + \frac{4EI\theta_C}{5} = -43.2 \rightarrow (b)$$

$$\theta_B = \frac{16.14}{EI} = 0.473$$

$$\theta_C = -\frac{62.07}{EI} = -54.23$$

(iv) Span AB

$$M_{AB} = -26.4335 \text{ KN-m.}$$

$$M_{BA} = 27.143 \text{ KN-m.}$$

Span BC

$$M_{BC} = -50.1136 \text{ KN-m.}$$

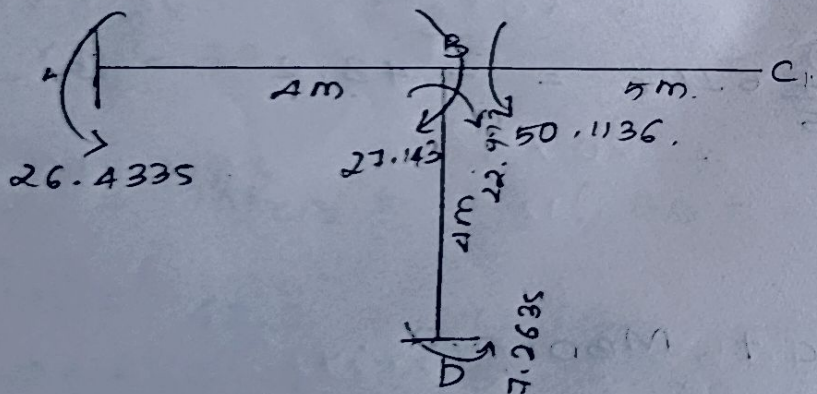
$$M_{CB} = 0$$

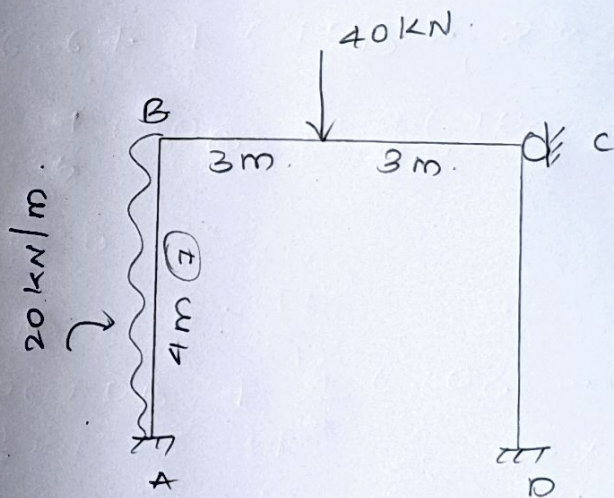
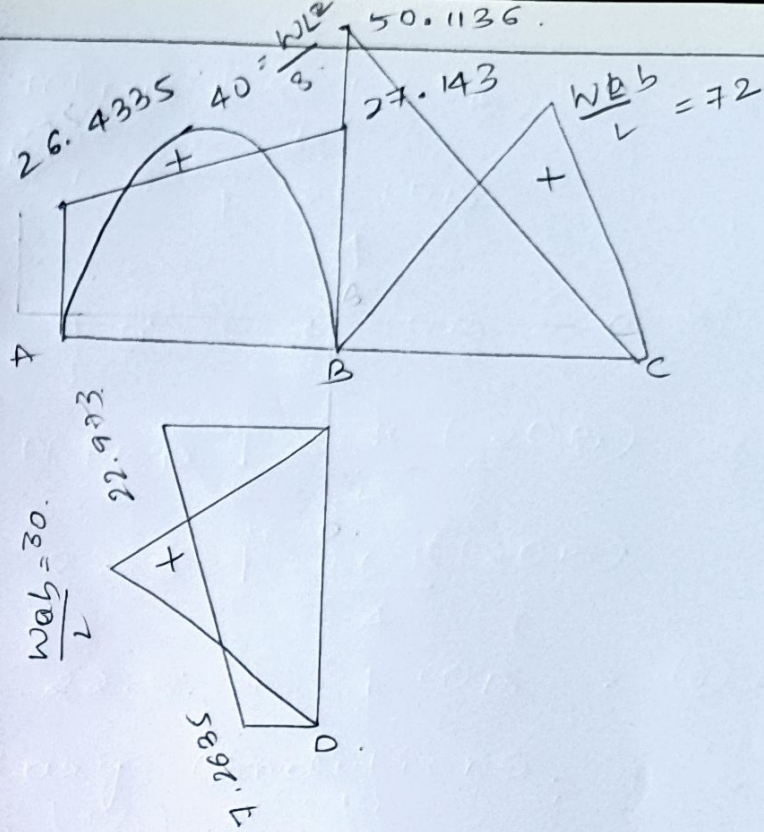
Span DB

$$M_{DB} = -7.2635 \text{ KN-m.}$$

$$M_{BD} = 22.973 \text{ KN-m.}$$

BMD.





Degree of freedom = 2 (i.e. θ_B and θ_C).

The given frame is non sway.

fixed end moments.

span BC.

$$M_{fBC} = -\frac{PL^2}{12} = -\frac{40 \times 6^2}{12} =$$